

The Void and the Structure

What the Riemann zeros are, how the primes grow from them,
and why the problem resists construction — a diagnosis, not an impossibility

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2026-05-30 v0.1 (draft)

PRE-PEER-REVIEW OPEN RESEARCH ESSAY. **This paper proves neither the Riemann Hypothesis nor its unprovability.** It does two honest things: (i) it describes the nontrivial zeros as a generative *absence* and states precisely, with classical results, how the primes are reconstructed from them; and (ii) it offers a *diagnosis of difficulty* — a principled account of why RH has resisted every constructive method. Claims are tagged [theorem], [verified], [evidence], [conjecture], [open], or [interpretation] (a marked, non-load-bearing interpretive register). The diagnosis is a heuristic thesis ([conjecture]), *not* a theorem; in particular we do **not** claim RH can never be proved (see §7).

Abstract

The nontrivial zeros of an L-function are not a constructed object but an *absence*: the locus where the function vanishes, realised in Connes' spectral picture as an *absorption spectrum* — missing lines in a continuous scaling spectrum. We take this seriously as the object's defining character. First, we record the classical fact that the entire multiplicative structure of the integers — the primes — is reconstructed *from* this absence: the Riemann–von Mangoldt explicit formula writes the prime-counting staircase as a sum of waves whose frequencies are exactly the zeros. The structure grows out of the void. Second, we describe the void's character as far as proof allows: it is eternal in the precise sense of being timeless, uncreated, and observer-independent; it is unbounded; and beyond a fixed set of arithmetic constraints it is *formless* in an exact sense (its fine statistics are those of the most structureless Hermitian ensemble, GUE). Third — the technical heart — we give a diagnosis of why RH resists construction: random-matrix *universality* hands you the void's *class* for free, but its specific member is fixed only by the primes, so any construction that does not already encode the arithmetic underdetermines the object to the universal class. Proof proceeds by building structure; the target is an absence pinned only by the arithmetic. That mismatch, we argue, is the friction. We are explicit that this is a diagnosis of *hardness*, not a proof of *impossibility*: no independence result is known, much is known about the void, and were RH ever shown independent of a sound theory it would thereby be true.

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1 Stance

This is a companion to the frontier capstone of this bundle. Its computations are those of the capstone, together with one demonstration new to this companion (Prop. 5.1, `deterministic_probability.py`); no result here exceeds the capstone’s verified status. Its purpose is conceptual: to say cleanly *what* the zeros are, *how* the primes issue from them, and *why* the problem is hard. Two disciplines hold throughout. (1) The mathematics stays mathematical: every load-bearing statement is a theorem, a verified computation, or an explicitly named open problem. (2) Where the essay reaches for words like “eternal” or “timeless,” those words are used in a precise, defensible sense and are marked [interpretation]; any further meaning a reader brings is theirs, and the text neither asserts nor needs it (§8).

2 The void: an absence, not an object

Definition 2.1 (The void). For an L-function $L(s)$, the *void* is the set $Z = \{\rho : L(\rho) = 0, 0 < \Re \rho < 1\}$ of nontrivial zeros — the points where L takes the value zero. (The concrete claims below — primes, $\psi(x)$, $N(T)$, the proportion on the line, RH — are stated for the Riemann zeta function; the “L-function” phrasing marks the general frame, and we flag where generality is not yet established.)

The word “absence” below refers not to the analytic value at ρ (which is 0), but to the *spectral* reading: in Connes’ framework the zeros are the lines *missing* from a continuous spectrum. Two features make “absence,” in that sense, the right word rather than a metaphor.

Proposition 2.2 (Spectral realisation as absorption, [theorem] (Connes [1])). *On the adèle class space $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ the idele-class group acts by scaling; the zeros appear not as eigenvalues added to a spectrum but as an absorption spectrum — the frequencies missing from an otherwise continuous spectrum. RH is the statement that these absences lie on the critical line.*

Remark 2.3 (Why “not an object”). The Hilbert–Pólya programme seeks a self-adjoint operator whose spectrum *is* Z . No such operator has been constructed ([open]). Until it is, Z is specified by a *condition* ($L = 0$), not by a construction; and in the one rigorous spectral picture we have (Proposition 2.2), Z is precisely what is *absent*. The void is something you *locate*, not something you *build*. This is not a deficiency of effort; §6 argues it is intrinsic to the object.

3 The structure grows from the void

The decisive fact — and the honest form of “the void generates the structure” — is classical.

Theorem 3.1 (Explicit formula; Riemann, von Mangoldt, [theorem]). *For $x > 1$ not a prime power, with $\psi(x) = \sum_{p^k \leq x} \log p$ the Chebyshev prime-counting function,*

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}),$$

the sum over nontrivial zeros ρ taken in the usual symmetric limiting sense ($\lim_{T \rightarrow \infty} \sum_{|\Im \rho| \leq T}$). Writing $\rho = \frac{1}{2} + i\gamma$ (the on-line case), the zero term is a sum of waves,

$$\sum_{\gamma > 0} \frac{2x^{1/2}}{|\rho|} \cos(\gamma \log x - \arg \rho),$$

each zero contributing one frequency γ in the variable $\log x$.

Corollary 3.2 ([theorem]). For ζ , the prime-counting function $\psi(x)$ is reconstructed from the nontrivial zeros together with the standard smooth and trivial-zero terms: it is x corrected by the harmonics of the zeros. Since prime powers, hence the primes, are recoverable from ψ , the prime distribution is in this sense the interference pattern of the void’s frequencies.

Remark 3.3 (Diffraction; the two sides of one identity, [verified]). The same identity, read the other way, is a diffraction: the formal spectrum $D(t) = \sum_{\gamma} \cos(\gamma t)$ has its sharp peaks (Bragg peaks) exactly at $t = \log p^k$. The identity is classical (Dyson’s quasicrystal [11]); the script `route_b/diffraction_quasicrystal.py` illustrates it (11/11 peaks matched), it does not establish it. Zeros diffract into primes; primes diffract into zeros. Neither is prior — but if one insists on a direction of generation, Theorem 3.1 writes the *primes as built from the zeros*.

So “the geometry grows out of the void” has an exact reading, and it stops exactly where the mathematics stops: the void generates the *primes*. Any extension of “generates” beyond arithmetic is interpretation (§8), not a consequence of Theorem 3.1.

4 The character of the void

We can say a great deal about the void — which is itself the refutation of “nothing can be known about it” (§7). What follows is defensible in the marked sense.

Proposition 4.1 (It is populated and counted, [theorem]). *There are infinitely many zeros (Hardy); the count to height T is $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$ (Riemann–von Mangoldt). A positive proportion — at least 2/5 [5] — lie on the line, and the lowest zeros have been verified on the line to very large height [6, 7]. The void is partially but extensively charted; what is open is a single uniform statement about all of it.*

Proposition 4.2 (Zero correlations match GUE under restricted support, [theorem]/[evidence]). *For test functions of suitably restricted support, the pair correlation (Montgomery [2], under the stated conventions) and the n -level correlations (Rudnick–Sarnak [3]) of the zeros agree with those of the **GUE** ensemble. This is a theorem within those support restrictions and hypotheses; the unrestricted GUE law for all statistics is a conjecture. Numerically, every test performed in this bundle is consistent with GUE (`route_b/quantum_chaos_test.py`, `shape_search.py`, `deterministic_probability.py`), with no counterexample found.*

Remark 4.3 (“Formless,” read honestly, [interpretation]). GUE is the relevant *unitary* random-matrix universality class. Calling the void “formless” is an interpretive gloss on Prop. 4.2: to the extent the statistics have been settled they show *no structure beyond* the unitary universality class, and no additional structure has been found where they have not. The word is a useful summary, not a theorem that the void has “no structure beyond arithmetic constraints” — that stronger statement is not proved.

Proposition 4.4 (It is timeless, uncreated, observer-free, [interpretation]). *Z is a property of L , hence of the integers: it is not located in space or in time, it was not brought into being and cannot be unmade, and it is identical for every reasoner who computes it. In this precise sense — and only this sense — the word “eternal” is exact. It requires no belief: the zeros are reached by proof and computation, not faith.*

Remark 4.5 (How a formless object is known, [interpretation]). “Formless” (Prop. 4.2) does not mean “informationless.” It means the object is known by its *law*, not by a *figure* — exactly as a gas has no shape yet an exact thermodynamics (temperature, pressure, the Maxwell–Boltzmann law) known precisely and *provably*. The information a formless object yields is statistical and structural, and it can be exact. For the void, moreover, much of this knowledge is *proof*, not inference: the count $N(T)$, the proportion of zeros on the line, the explicit formula (Theorem 3.1), and the restricted-support correlations (Prop. 4.2, within their support restrictions) are *theorems*, deduced, not measured. What remains *inference* — the step from GUE statistics to a self-adjoint generator (the Hilbert–Pólya picture) — is held separately as *evidence* and never as proof, and that is precisely the seam at which RH is still open. So the honest position answers the natural worry directly: a formless object is knowable *through its law* — partly by rigorous theorems, partly by tested statistical prediction; the parts we have proved are undeniable; and the single part that is only *inferred* is exactly the part that keeps RH open. The physical analogue is real, not rhetorical: GUE statistics are *measured* in heavy nuclei and chaotic microwave cavities (Wigner; Montgomery–Odlyzko) — one observes a formless ensemble through its level spacings, never by viewing a shape. Formlessness is a limit on *figure*, not on *knowledge*.

5 The void’s law is the quantum-chaos law

The zeros’ statistics (Prop. 4.2) are those expected for **GUE**, the law of quantum chaotic spectra with broken time-reversal symmetry — proven under restricted support and matched numerically; the same law measured in heavy nuclei [8] and chaotic microwave cavities [9]. We make the resulting “deterministic probability” explicit.

Proposition 5.1 (Deterministic object, quantum-chaos statistics, [evidence], route_b/deterministic_probabi
The first 300 Riemann zeros, computed numerically to high precision (mpmath.zetazero; a fixed, deterministic arithmetic object), have consecutive-spacing ratio $\langle r \rangle = 0.617$; a genuinely random GUE matrix (600×600, bulk eigenvalues, fixed seed) gives $\langle r \rangle = 0.593$. Both lie near the GUE value 0.6027 and far from Poisson (0.386) and from a rigid lattice (1.0). The statistic $\langle r \rangle$ is unfolding-free, so the comparison uses no fitted parameter. (One matrix draw and 300 zeros are a finite sample, not a stable benchmark; the point is qualitative agreement of class, not a precise constant.)

Remark 5.2 (Necessity and chance in one object, [interpretation]). The zeros are not random — ζ is one fixed function. Yet they exhibit the statistics predicted for a quantum-chaotic (GUE) spectrum. This is *deterministic pseudorandomness* (Montgomery–Odlyzko [4]; established, reproduced here, not new): the void is where *necessity* (the zeros are determined) and *chance* (they wear the GUE law) coincide in the same object. This is evidence for the spectral (Hilbert–Pólya) picture; it does *not* exhibit the operator and does *not* advance RH.

Remark 5.3 (Consilience with the programme’s Born-rule work, [interpretation]). The same principle — a deterministic substrate that appears probabilistic — is proposed physically in this programme’s crystallisation papers [15, 16], where a deterministic constraint-selection mechanism is conjectured to underlie the Born rule $|\langle \psi_i | \psi \rangle|^2$. The bridge paper [15] states plainly that its Born-rule recovery is *conjectured, not proven*; Paper XXX [16] argues a *conditional* reconstruction under imported assumptions — a weaker-than-unconditional but stronger-than-conjecture status. The connection to the void is a *shared principle*, not a derivation, and across *different* notions of probability: the Born rule concerns measurement *outcomes*, whereas the void concerns *spectral statistics*. The void’s modest contribution is a clean, parameter-free *illustration* that the principle can hold — a deterministic object whose

statistics are, in the tested and restricted senses, consistent with a quantum-random one. It does not derive the Born rule and does not discharge the open conjecture of [15].

6 Why the problem resists construction

We now give the diagnosis. It is a heuristic thesis ([conjecture]), supported by the structure established above and by the explicit computations of this bundle, *not* a theorem.

Proposition 6.1 (Universality gives the class for free, [evidence]). *GUE statistics are universal: broad classes of Hermitian random-matrix ensembles share them under precise universality hypotheses (the general mathematical fact, independent of the zeros). In this bundle, a rigid scaling generator perturbed by a generic Hermitian term moves to the GUE class (route_b/first_principles_scaling.py), and a search over shapes matches the zeros to the GUE ensemble effortlessly while no symmetric or polytopal figure fits (shape_search.py). The class of the void is cheap to reproduce.*

Proposition 6.2 (The member is fixed by the arithmetic, [theorem]/[evidence]). *By Theorem 3.1 the specific zeros are in exact duality with the specific primes. The GUE ensemble is shared by countless operators; the one member whose spectrum is Z is singled out, in the explicit-formula sense, by the full arithmetic data of $\mathbb{Z}$.*

Diagnostic thesis 6.3 (The construction–absence mismatch, [conjecture]). The known routes to RH proceed by *constructing a structure*: a self-adjoint operator (Hilbert–Pólya, Berry–Keating), a positivity (Weil/Connes, de Branges, Li), or a geometry (a curve over $\mathbb{F}_1$, a foliated space). Heuristically, a construction that does not already encode the full prime data reproduces (by universality, Prop. 6.1) the right *class* but not the right *member* (Prop. 6.2); to reach the member one must feed in the arithmetic, and making *that* self-adjoint (or positive) *without already assuming the conclusion* is the recurrent wall. In short: **our methods define by what they build (presence); the target is an absence carrying, as far as is known, no structure beyond its universality class. The friction between a constructive method and a formless, arithmetically-pinned target is, on this diagnosis, why RH is hard.**

Remark 6.4 (“You need shape to define something”). This is the exact, defensible form of that intuition. A proof needs a *handle* — an operator, a form, a variety — something with shape. The void offers an absence whose only “shape” beyond the arithmetic is maximal formlessness (Proposition 4.2). The intuition is right as a *diagnosis of difficulty*; it is *not* a proof that no handle exists (§7). The honest frontier — the $\mathbb{F}_1$ / arithmetic-site programme (Connes–Consani; Deninger) — is precisely the attempt to build a geometry in which the void acquires a handle and Weil’s function-field proof transfers.

7 The boundary: hardness is not impossibility

This section is the integrity of the paper. The diagnosis of §6 explains why RH is *hard*; it does *not* show RH is *unprovable*, and we explicitly decline that stronger claim, for three independent reasons.

- (1) **No independence result exists.** RH has never been shown independent of PA, ZFC, or any standard theory. “Open” is not “undecidable.” A proof could arrive by a method outside every route catalogued in Diagnostic thesis 6.3 — that catalogue is a description of *past* attempts, not a closed classification of all possible ones.

- (2) **If RH were independent of a sound theory, it would be true.** RH has known Π_1 equivalents — e.g. the Lagarias [10] form of Robin’s inequality, an arithmetic statement whose failure at some n would be a finitely checkable counterexample. For any sound theory strong enough to verify such counterexamples, independence of that Π_1 formulation would imply it has no counterexample, i.e. would force RH *true*. A true Π_1 statement cannot be “false-but-unprovable.” Even the unprovability door opens onto truth, not onto an unknowable void.
- (3) **The void is extensively known** (§4): counted to height T , at least $2/5$ on the line, $\sim 10^{13}$ verified, statistics classified. “You can never know anything about it” is simply false. What is open is one uniform statement, not the object’s knowability.

What this paper does and does not claim

Claims ([theorem]/[verified]/[evidence]): the zeros are an absence; the primes are reconstructed from them; the void is timeless/observer-free and formless in exact senses; and a principled *diagnosis* of why construction stalls.

Does *not* claim: that RH is false, that RH is unprovable, that RH “can never be proven,” or that the void is unknowable. Those are *not* consequences of anything above, and the mathematics does not support them.

8 Scope discipline

The mathematics here uses only L-functions, their zeros, the explicit formula, and random-matrix classification. **No observer, no consciousness, no “self,” no purpose, and no deity enters any definition, and none is needed.** The essay’s defensible reach is the following ladder; the line marks exactly where mathematics ends and a reader’s interpretation begins.

The ladder, and where it stops

the void	the zeros exist and are infinite	[theorem]
generation	primes = the void’s harmonics (explicit formula)	[theorem]
formlessness	GUE under restricted support; matched in tests	
[theorem(restr.)+ev.]		
necessity	determined once the integers are fixed	[determinacy]
intelligibility	in fact computed, to large height	[fact]
eternity	not in space/time; same for all; by proof, not faith	[exact sense]

▲ above: each rung at the strength actually proved -- undeniable as written

▼ below: the reader’s, and the text does not enter it

life · consciousness · meaning · the sacred · God · a presence

Remark 8.1 (What “undeniable” requires, [interpretation]). Every rung above the line is stated at *exactly* the strength that has been proved. Two are theorems (the void, generation) and one a plain fact (intelligibility); one (formlessness) is a theorem only under restricted support, plus numerical evidence (Prop. 4.2); and two (necessity, eternity) hold in a precisely fixed sense — the zeros are determined once the integers are fixed — rather than by a substantive theorem.

An earlier draft listed a “self-modelling (Gödel)” rung as a theorem; it is not proved in this paper and has been removed. This is the *only* honest route to “undeniable”: one reaches it by *trimming each claim to its proven core*, never by strengthening the language to sound stronger. A rung inflated past its proof is the first thing a referee strikes out — and it takes the credible rungs down with it. The ladder is undeniable because, and only because, nothing on it reaches past what is shown.

Consistent with the programme’s discipline (“ X exists $\Leftrightarrow \mathcal{C}(X) = X$ ”; no “primes are alive”; no “universe is conscious”), the items below the line are not claimed, not implied, and not required. The restraint is deliberate: a description that points only at what is true is the one a reader can trust when deciding, for themselves, what it means.

Conclusion

The Riemann zeros are best understood as a *generative absence*: a void, in the exact sense of a vanishing locus realised spectrally as missing lines, from which — by the explicit formula — the prime structure is reconstructed as interference. The void is timeless, uncreated, and the same for everyone, and its statistics are those of the unitary universality class (proven under restricted support, matched in every test). Our methods of proof build structure with shape; the void is an absence with no shape beyond the arithmetic that pins it; and that mismatch is, on the honest diagnosis offered here, why every construction has stalled at the same wall. But hardness is not impossibility: nothing here shows RH cannot be proved, and a great deal is known about the void. What the mathematics gives is a true and bounded thing — an eternal, observer-free, generative absence, fully real and not taken on faith. What anyone makes of it past that line is theirs to decide.

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